SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I SK.Arshad Basha(192312376) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

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CO1-AT3 (Descriptive Test)

Type of Intelligent Agents in smart Home Automation

Simple Reflex Agent

works on condition-action rules.

uses only current input, no memory

In the system

If motion detected → turn on lights.

If temperature rises → turn on AC.

If door opens at night → trigger alarm.

2) model-Based Agent

* maitains an internal state (memory) of the environment

In the system

* Remembers which rooms are the occupied.

* Tracks time is more (day/night).

3) Goal-Based Agent

Takes action to achieve specific goals

In the system

* maintain comfortable temperature

* Ensure home seurity.

* Decides action like a adjusting AC.

Analysis:-

* more flexible and intelligent.
* can evaluate different. actions.

④ Utility-Based Agent

* use a utility function to measure:
* choose action that maximize

Balances:-

* comfort (lighting, temperature)
* Engergy efficiency
* security

Example:-

Reduce AC slightly to save energy without affecting confort

Best Approach (Justification)

A hybrid approach is most effective.

Advantages

* optimizes multiple objectives
* Learns user preferences
* most intelligent among the four agent types

Disadvantages

* complex to design
* Requires more computation and data.

Conclusion

A combination of model-based, goal-based, and the utility based agent with reflex rules is ideal

* comfort through personalized.
* Energy effeciency ~~by~~ reducing unnessary usage.

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2) sol:- Robot in unknown maze - uninformed search strategies

A robot is an unknown maze must explore paths a step-by-step with out prior knowledge. uniformed search algorithm like uniform cost (search (UCS). and the Iterative Deepening search (IDS)

* ① Uniform cost search (UCS):-

* Expands the node with the lowest path of the cost.

* It is uses a priority queue.

In the maze:-

* Robot explores path based on minimum travel cost.

* Alway guarantees the least-cost path to the exit.

Examples:- ① Distance

② Time

Advantages

* complete (will find a solution if it is exists).
* optimal (finds lowest-cost path).

DisAdvantages

* High time complexity:- $O(b^d)$ $O(b^{d})$
* It can explore many nodes.

Iterative Deepening search (IDS)

* combines Depth-first search + Breadth-First search
* Repeats with increasing depth limits.

In the maze:-

* Robot explores the path up to depth 1, then 2, then 3.
* Eventually finds the exist at shallowest level

Advantages

* complete
* optimal (for equal step)

Dis Advantages

* Repeats exploration → higher time
* ~~Sa~~ Not suitable when costs vary.

Comparision (Time & space)

Alogrithm	Time complexity	Space complexity	optimal
~~Ches~~ UCS	High	Very High	Yes
IDS	moderate High	Low	Yes (unit cost)

Complexity

Time complexity :- $O(b^d)$

Space complexity :- $O(bd)$

where;

b = branching factor

d = depth of the solution

① simple Reflex Agent

* Acts only on current preception

* cannot handle maze

② model - Based Agent

* maintains internal map of explored path.

* useful for avoiding revisiting same location

③ Goal - Based Agent :-

* Has goal : reah exist

* uses search algorithms (UCS/IDS) to plan path

④ utility Based Agent

* Evalute s best path on cost, Risk.

chooses most effecient solution.

clusion:-

* UCS is best optimal path finding but uses more memory.

* IDS is best memory efficiency but a less cost aware.

* It is combines the memory, planning and the optimal path sleetion, enabling the robot to navigate efficiently.

* Thus a hybrid intelligent agent UCS provides the accuray, efficiency and decision making in the unknown maze.